STAT 527 Advanced Regression Analysis
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Properties of Least Squares
(Part II: Gauss-Markov Theorem)

Jingbo Liu

University of Illinois, Urbana-Champaign

Is Least Squares Estimator Optimal?

We chose to use the least squares minimization to estimate model parameters.

- A crucial question is - "Can some other approach do better than least squares to estimate the regression coefficients for the linear model?"
- What do we mean by better performance for an estimator?

Gauss-Markov Conditions

Let's look at some conditions under which $\hat{\boldsymbol{\beta}}$ is an optimal estimator.
the **Gauss-Markov (G-M) conditions**:

$$GM1: \quad E[\boldsymbol{\varepsilon}] = \mathbf{0}$$
$$GM2: \quad E[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T] = \mathrm{var}(\boldsymbol{\varepsilon}) = \sigma^2 \boldsymbol{I}$$

(These are the minimal assumptions that we have been making about the errors.)

Gauss-Markov Theorem

Under the G-M conditions it can be shown that $\boldsymbol{c}^T\hat{\boldsymbol{\beta}}$ is the **best linear unbiased estimator** (BLUE) of the parameter $\boldsymbol{c}^T\boldsymbol{\beta}$ for all $\boldsymbol{c}$: It has the smallest variance among all linear, unbiased estimators.

Theorem (Gauss-Markov):

Let $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{y}$ and $\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$.

Then under conditions $GM1$ and $GM2$, the estimator $\boldsymbol{c}^T\hat{\boldsymbol{\beta}}$ of the estimable parameter $\boldsymbol{c}^T\boldsymbol{\beta}$ is the BLUE.

estimable just means that an unbiased estimator exists

Proof of Gauss-Markov

Proof: Let $\boldsymbol{a}^T\boldsymbol{y}$ be another unbiased estimator of $\boldsymbol{c}^T\boldsymbol{\beta}$. Since $\boldsymbol{a}^T\boldsymbol{y}$ is unbiased,

$$\boldsymbol{c}^T\boldsymbol{\beta} = E[\boldsymbol{a}^T\boldsymbol{y}] = \boldsymbol{a}^T\boldsymbol{X}\boldsymbol{\beta}$$

for all $\boldsymbol{\beta}$, and so $\boldsymbol{a}^T\boldsymbol{X} = \boldsymbol{c}^T$.

Proof Continued

By standard calculations we know that

$$\mathrm{var}(\boldsymbol{a}^T\boldsymbol{y}) = \sigma^2\boldsymbol{a}^T\boldsymbol{a}$$

and

$$\mathrm{var}(\boldsymbol{c}^T\hat{\boldsymbol{\beta}}) = \sigma^2\boldsymbol{c}^T(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{c}$$
$$= \sigma^2\boldsymbol{a}^T\boldsymbol{X}(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T\boldsymbol{a} = \sigma^2\boldsymbol{a}^T P_X\boldsymbol{a},$$

where

$$P_X = \boldsymbol{X}(\boldsymbol{X}^T\boldsymbol{X})^{-1}\boldsymbol{X}^T$$

is the projection matrix onto the column space of $\boldsymbol{X}$.

Proof Continued

$$\mathrm{var}(\boldsymbol{a}^T\boldsymbol{y}) - \mathrm{var}(\boldsymbol{c}^T\hat{\boldsymbol{\beta}}) = \sigma^2\left[\boldsymbol{a}^T\boldsymbol{a} - \boldsymbol{a}^T P_X \boldsymbol{a}\right] = \sigma^2 \boldsymbol{a}^T\left[\boldsymbol{I} - P_X\right]\boldsymbol{a}.$$

This is a quadratic form with the matrix $\boldsymbol{I} - P_X$, so to show that $\mathrm{var}(\boldsymbol{a}^T\boldsymbol{y}) - \mathrm{var}(\boldsymbol{c}^T\hat{\boldsymbol{\beta}}) \geq 0$, we only must show that

$$\boldsymbol{M} = \boldsymbol{I} - P_X$$

is positive semi-definite (psd).

M is symmetric and is a projection matrix onto the null space of X. Recall that a projection matrix M satisfies $M^2 = M$. This proves that M is psd.

Properties under Normal Errors

- For Gauss-Markov, we did not assume errors to be normally distributed.
- With the additional assumption of normality of the errors, errors will not only be uncorrelated but will also be independent. Why?
- In this case,

$$\hat{\boldsymbol{\beta}} \sim N(\beta, \sigma^2(\boldsymbol{X}^T\boldsymbol{X})^{-1}); \quad (n-p')\,\frac{\hat{\sigma}^2}{\sigma^2} \sim \chi^2(n-p').$$

- Also, $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$ are independent!

Some Key Properties of Normal Distribution

Suppose $Z \sim N(\mathbf{0}, I_{m\times m})$, that is, its components are $i.i.d.$ standard normal.

- For a non-random matrix A, $AZ \sim N(\mathbf{0}, AA^T)$.
- If A and B are non-random matrices, then AZ and BZ are independent if and only if $AB^T = 0$.
- For an orthogonal projection matrix $P_{m\times m}$, $Z^T PZ$ follows a Chi-squared distribution with degrees of freedom $=$ rank of P (same as trace of P).

These properties can be used to establish the results on the previous slide.

Remarks on the Assumptions

- Gauss-Markov Theorem assumes an unbiased estimator is linear, and the noise is zero mean, uncorrelated, and homoscedestic.

- We can drop the linearity assumption but assumes unbiasedness for *all* such noise distributions; see

Bruce E. Hansen, A Modern Gauss-Markov Theorem, 2020.

- We can drop the linearity assumption but assumes Gaussianity of the noise (Cramer-Rao bound)

Rao-Cramer Lower Bound

Let $\hat{\theta}$ be an unbiased estimator of θ. Then, the Rao-Cramér Lower Bound is given by:

$$\text{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)}$$

where $I(\theta)$ is the Fisher Information of θ.

- Provides a benchmark for the efficiency of an estimator.
- If an estimator achieves this bound, it's considered efficient.